

EVALUATING PREDICTION PROCEDURES IN MULTIVARIATE REGRESSION:

A RESAMPLING APPROACH

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Summary: The paper reviews a few methods of predicting several response variables in multivariate linear regression. Several tools based upon validation and resampling are provided to compare these methods. These decision making tools are illustrated using two practical examples.

1. INTRODUCTION

The number of approaches available for the prediction of more than one response variable in linear regression has increased in the recent literature. Selection of variable techniques have been discussed by McKay (1977, 1979), van der Merwe, *et al.* (1983) and Sparks, *et al.* (1983, 1985). The generalisation of the Stein-type shrinkage estimator to multivariate regression has been covered by Sclove (1971), Zidek (1978), van der Merwe and Zidek (1980) and Groenewald, *et al.* (1982). The ridge regression techniques first introduced by Hoerl and Kennard (1970) have been extended by

Sclove (1973) and Brown and Zidek (1978).

A review of some of the above techniques is carried out. Also covered are two additional methods, i.e. a special case of Zellner's (1962) seemingly unrelated regression equations (SURE) and to shrink the ordinary least squares (OLS) estimate of selected regression coefficients. These will be discussed and compared in terms of their mean square error of prediction.

2. THE MODEL AND PREDICTION PROCEDURES

Consider the familiar multivariate regression model

$$(1) \quad Y = X\beta + E$$

where: Y is an nxq matrix of n observations on q responses;

X is an nxk matrix of n observations on k predictors;

β is a $k \times q$ matrix of regression coefficients; and

E is the nxq matrix of random errors.

Assume that E is distributed with mean equal to the zero matrix and covariance matrix $I_n \otimes \Sigma$. The future value of vector y of responses, at a known value of the predictor variables (say x_0), is predicted as follows:

$$(2) \quad \hat{y}' = x_0' \hat{\beta}$$

where $\hat{\beta}$ is some estimator of the regression coefficients. The efficiency of $\hat{y}$ depends on the estimate $\hat{\beta}$. The OLS estimate of β is given by

$$(3) \quad \hat{\beta}_O = (X'X)^{-1}X'Y.$$

If $X'X$ and Σ are diagonal matrices then estimator (3) will generally be good. However, $\hat{\beta}$ is adversely affected by two types of instabilities, namely the effects of collinearities amongst the predictor variables and what van der Merwe and Zidek (1980) called "noisy" Y-canonical variates. The former instability can have a severe effect on estimator (3). It can cause (say) the j-th row of $\hat{\beta}$ to have artificially large values or an element of $\hat{\beta}$ to take on the wrong sign (Hoerl and Kennard (1972)). If, in the presence of "noisy" Y-canonical variates, practitioners use estimator (3) then they are failing to take cognizance of the information available in the off diagonal elements of Σ . Sclove (1971), the first to account for the information in Σ , extended James and Stein's (1961) estimator. Since then, other estimators which make use of the information in Σ have followed.

The multivariate shrinkage estimator takes the form:

$$(4) \quad \hat{\beta}_S = \hat{\beta}_O F$$

where $0 \leq F \leq I_q$. The following estimates of F will be examined:

$$F = G \text{ if } 0 < G \leq I_q \text{ and } F = 0 \text{ if } G < 0$$

where

$$(a) \quad G = I_q - \alpha_I (\hat{\beta}_O' X' X \hat{\beta}_O)^{-1} S \quad (\text{Zidek (1978)});$$

$$\alpha_I = (n-k) \max(0, k-q-1) / (n-k+q+1) \text{ and}$$

$$(n-k)S = Y'Y - \hat{\beta}_O' X' Y;$$

$$\begin{aligned}
 \text{(b)} \quad G &= I_q - \alpha_2 (\hat{\beta}'_o \hat{\beta}_o)^{-1} S && \text{(Brown and Zidek (1980));} \\
 \alpha_2 &= \max (0, \text{tr}(X'X)^{-1} - (q+1)\lambda_k^{-1}) \quad \text{and} \\
 \lambda_k &\text{ is the smallest eigenvalue of } X'X.
 \end{aligned}$$

These shrinkage estimators make use of the information available in Σ but are not designed to control any instability caused by collinearities amongst the predictor variables.

The adaptive multivariate ridge regression estimator (Sclove (1973), Brown and Zidek (1978, 1980)) is an estimator which could control both instabilities mentioned earlier. This estimator is denoted by $\hat{\beta}_o(K)$ where K is the $q \times q$ ridge matrix. Let $\hat{\beta}_o(K)$ be the $kq \times 1$ vector formed by stacking the columns of $\hat{\beta}_o(K)$, then

$$(5) \quad \hat{\beta}_o(K)' = (I_q \otimes X'X + K \otimes I)^{-1} (I_q \otimes X'X) \hat{\beta}_o$$

where

$$\begin{aligned}
 \hat{\beta}_o' &= (\hat{\beta}'_{o1} \dots \hat{\beta}'_{ok}), \quad \hat{\beta}_{oj} \text{ denotes the } j\text{-th column of } \hat{\beta}_o \\
 &\text{and } \otimes \text{ denotes the usual Kronecker product.}
 \end{aligned}$$

Brown and Zidek (1978) examined a number of possible forms for the ridge matrix K . In order to limit their numerous options, only the following K matrices are considered (see Brown and Zidek (1980)):

(i) The modified Hoerl, Kennard and Baldwin (1975) (H-K-B) rule:

$$K = (k-q-1)(\hat{\beta}'_o \hat{\beta}_o)^{-1} S;$$

- (ii) Sclove's (1973) rule: $K = (k-q-1)\bar{\lambda}(\hat{\beta}_0'X'X\hat{\beta}_0)^{-1}S$ and
 $\bar{\lambda} = \text{tr}(X'X)/k.$

If the variables are all standardized before the analysis, then the latter rule is identical to the multivariate extension of Lawless and Wang's (1976) rule, i.e. $K = (k-q-1)(\hat{\beta}_0'X'X\hat{\beta}_0)^{-1}S$. Sclove's rule will adjust with a change in scale of the predictor variables while Lawless and Wang's rule does not. Hence, if the practitioner does not standardize, then the author suggests Sclove's rule should be given preference over the extension of Lawless and Wang's rule.

Although van der Merwe and Zidek (1980) suggested that an optimal rule for estimating β may well be an adaptive ridge rule, it will be demonstrated that ridge rules are not always optimal in terms of prediction.

In a subset of $k-p$ predictor variables (say X_{out}) are almost linearly related to the remaining p predictor variables (say X_{in}) then it may be advantageous to constrain the regression coefficients, relating to X_{out} , to equal zero and then find the OLS estimate of β_{in} (the regression coefficients corresponding to X_{in}). That is, estimate β as follows:

$$(6) \quad \beta_{sr} = \begin{bmatrix} \hat{\beta}_{in} \\ 0 \end{bmatrix} \quad \text{where} \quad \hat{\beta}_{in} = (X_{in}'X_{in})^{-1}X_{in}'Y$$

(no loss of generality in assuming $X = [X_{in} X_{out}]$). The performance of prediction using $\hat{\beta}_{sr}$ depends on the appropriateness of the split of X into X_{in} and X_{out} . This is often done via variable selection techniques where estimation and fitting are based on the same data set. Such empirical selection techniques

can lead to models being chosen which are less efficient for prediction than the full model. If predictor variables are seriously collinear, then the chances are, that empirical selection procedures will choose a subset of predictor variables which are less collinear, but this is at the risk of choosing the wrong predictors.

$\hat{\beta}_{sr}$ used for prediction may control the problems that collinearities amongst the predictor variables cause, however it does not make use of the information in Σ . In order to control both instabilities mentioned earlier in the paper, the following shrinkage of $\hat{\beta}_{sr}$ is considered:

$$\hat{\beta}_{ssr} = \hat{\beta}_{sr} F$$

where:

F is the shrinkage matrix as defined before.

If a different subset of predictor variables is selected for each response then Zellner's (1962, 1963) SURE may prove useful (see Sparks (1987)). Let $X_{in}(j)$ and $\beta_{in}(j)$ be the corresponding subsets of X and β_j (the j -th column of β) respectively chosen as "best" for prediction of j -th response. Define

$$Z_{in} = \begin{bmatrix} X_{in}(1) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & X_{in}(q) \end{bmatrix}; \quad S_c = S \otimes I_n;$$

$\beta'_{glS} = (\beta'_{in}(1) \dots \beta'_{in}(q))$ and $y_c = (y'_1 \dots y'_q)$
where y_i is the i -th column of Y .

The generalised least squares estimate (glS) of the selected regression coefficients is given by

$$(7) \quad \hat{\beta}_{gls} = (Z'_{in} S_c^{-1} Z_{in})^{-1} Z'_{in} S_c^{-1} y_c.$$

The constrained gls estimator of β is $\hat{\beta}_{gls}$ which defines its i -th row and j -th column element (denoted $\hat{\beta}_{gls}(i,j)$) as follows:

$$\begin{aligned} \hat{\beta}_{gls}(i,j) &= \text{the corresponding element of } \hat{\beta}_{in}(j) \\ &\quad (\text{in } \hat{\beta}_{gls}) \text{ if the } i\text{-th predictor is} \\ &\quad \text{included in } X_{in}(j) \\ &= 0 \text{ otherwise.} \end{aligned}$$

Prediction using $\hat{\beta}_{gls}$ may go some way to excluding unnecessary predictor variables and also makes use of the information available in Z .

Experimenters are faced with these (and possible several other) alternative prediction methods and they have no means of finding which estimator "best" suits a particular practical example. In order to provide experimenters with several tools to help them resolve procedural choices, a decision forming process based on resampling methods and validation is presented. Validation techniques have been discussed and motivated in the more recent literature for multiple linear regression by Snee (1977), Rutherford (1981) and Copas (1983). These techniques will now be discussed in terms of comparing prediction methods.

3. METHODS OF COMPARING PREDICTION PROCEDURES

Copas (1983) defines "retrospective fit" as the fit to the original data used to estimate the regression coefficients and "prospective (or validation) fit" as the fit to new data. Since any assessment of "retrospective fit" uses the data "more than

once", it will often reflect an optimistic estimate of prospective fit. Prediction of future response values is more closely related to prospective fit rather than retrospective fit. Hence the comparison of prediction procedures should be based on a measure of prospective fit. Copas concentrated on maximizing the correlation between the observed and predicted value of the responses as a measure of prospective fit. This paper will concentrate on minimizing the sum of squared distances between the observed and predicted value. This measure will be denoted as the Validation Error Sum of Squares (or VESS).

The VESS will be computed as follows:

STEP 1: The construction sample is used to estimate the regression coefficients. Let this estimator be $\tilde{\beta}$.

STEP 2: The coefficients obtained in Step 1 are used to estimate the responses for each value in the validation sample i.e. denote y_{ij}^* as the i -th observation of the j -th response with corresponding predictor values $\mathbf{x}_i^*$ then $\tilde{y}_{ij}^* = \mathbf{x}_i^{*'} \tilde{\beta}_j$, where $\tilde{\beta}_j$ is the j -th column of $\tilde{\beta}$ obtained in Step 1.

STEP 3: VESS for the j -th response can be calculated by finding the sum of squared deviations of the observed value from the predicted value for all data points in the validation sample i.e.

$$\text{VESS}_j^m = \sum_i (y_{ij}^* - \tilde{y}_{ij}^*)^2.$$

For response j (and each particular construction and validation sample) the methods of prediction can be compared by examining VESS_j^m for all methods m . The ranking of the VESS_j^m

values reflects the orderings of the procedures. The method with the smallest $VESS_j^m$ -value is assigned a rank of one and is most preferred by the data.

The choice of method based on a single validation sample can lead to an incorrect ranking of the methods for prediction. The alternative is to compare methods over several splits of the data; one part being the construction sample and the other, the validation sample.

A well-known technique which may give reasonable estimates for $VESS_j^m$ is that based on the cross validation principle.

3.1 Cross validation

Assume there are n observations available. If selecting a validation sample of size n^* , then there are $t = \binom{n}{n^*}$ possible different splits of the n observations into a construction sample of size $n - n^*$ and a validation sample of size n^* .

Define $VESS_j^m(i)$ for the i -th split as above, then the standard error for prediction of the j -th response could be estimated by

$$\sqrt{\frac{t}{t-1} \sum_{i=1}^t VESS_j^m(i) / n^* t}$$
 where $VESS_j^m(i)$ is the VESS of the j -th response using method m for the i -th possible split of the data.

Additional information, for comparing methods, can be obtained by:

- (i) examining the empirical distribution of $VESS_j^m$. For each method, the empirical distribution of $VESS_j^m$ will be

approximately $\sigma_j^2 \chi_{n^*}^2(\lambda_m)$, where $\chi_{n^*}^2(\lambda_m)$ is a chi-squared distribution with n^* degrees of freedom and noncentrality parameter λ_m , and σ_j^2 is the variance of the predictions of response j made using method m . The best method will be the one which has the smallest noncentrality parameter. The value λ_m is a measure of all the biases involved using method m to predict the j -th response values.

- (ii) For each split of the data, the $\text{VSS}_j^m(i)$ values can be used to rank all methods, from "best" to "worst". The frequency distribution of these rankings could be a useful decision tool in comparing prediction methods.

An alternative resampling method receiving considerable attention in recent literature is the bootstrap (Efron (1982)).

3.2 Bootstrap validation error sum of squares

Bootstrap procedures can be handled either parametrically or nonparametrically. The nonparametric bootstrap makes no assumptions about the overall distribution of the data, however the original data set is regarded as a representative sample from a population or is representative of the area of the population within which prediction is desired. Available observations are known as the bootstrap population. A bootstrap sample will be a 'with replacement simple random sample' taken from the bootstrap population.

The parametric bootstrap assumes that the data follows some known distribution i.e. $[x'y']$ is assumed to have a distribution G^* with mean $[\bar{x}'\bar{y}']$; $\bar{x}$ and $\bar{y}$ are the sample mean vectors for

the predictor and response variables respectively; and covariance matrix

$$S^* = \begin{bmatrix} S_{xx} & S_{xy} \\ S_{yx} & S_{yy} \end{bmatrix}; \quad S_{xy} = X'(I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n') Y / (n-1) \quad \text{with } X$$
 and Y defined by (1) and $\mathbf{1}_n$ is an $n \times 1$ vector of ones (S_{xx} , S_{yx} and S_{yy} are similarly defined).

The population generated by G^* with mean and variance covariance matrix as mentioned above will be known as the bootstrap population. The bootstrap sample will be a random sample of observations generated from G^* (which is usually taken as normal)

(a) Nonparametric bootstrap

A bootstrap sample of $n_c (\leq n)$ observations is drawn with replacement, at random, from the bootstrap population. These n_c observations define the construction sample. The validation sample is made up of any observation in the bootstrap population which is not a member of the construction sample. Note that the validation sample size is not constant for each bootstrap sample. If $\tilde{y}_{jk}^*(i)$ denotes the predicted value of the k -th observation for the i -th response in the validation sample of the i -th bootstrap and $y_{jk}^*(i)$ is the observed value, then

$$(8) \quad \text{VESS}_j^m(i) = \sum_{k=1}^{n_i^*} (y_{jk}^*(i) - \tilde{y}_{jk}^*(i))^2$$

is defined as the bootstrap validation error sum of squares (n_i^* is the validation sample size for the i -th bootstrap).

The degrees of freedom of $\text{VESS}_j^m(i)$ may vary from one bootstrap sample to another. Hence it is suitable to examine the empirical distribution of the standardized statistic

$$V_j^m(i) = \frac{\text{VESS}_j^m(i) / \hat{\sigma}_j^2 - n_i^*}{\sqrt{2n_i^*}}$$

where: $\hat{\sigma}_j^2$ is the variance of the j -th response in the bootstrap population. The advantage of the above standardization is that, if there are no biases present in the prediction method then the mean of the empirical distribution of $V_j^m(i)$ will be zero. Any movement of this empirical distribution away from zero is related to λ_m , the noncentrality parameter. The empirical distribution of $V_j^m(i)$ can be compared to each m , the best method ' m ' is the one with the mean for $V_j^m(i)$ closest to zero and smallest variation from this mean.

The standard error for prediction can be given as

$$(9) \quad \text{SDV} = \sqrt{\frac{B}{\sum_{i=1}^B} \frac{n_i^*}{\sum_{k=1}^B} (y_{jk}^*(i) - \tilde{y}_{jk}^*(i))^2 / \sum_{i=1}^B n_i^*}$$

where: B is the number of bootstrap samples considered.

The frequency distribution of the rankings can be carried out as was described before.

(b) Parametric bootstrap

A bootstrap sample of size $n_c + n_v$ is generated from the bootstrap population with distribution defined by G . This sample is randomly split into a construction sample of size n_c and a

validation sample of size n_v . The comparison of the methods is identical to those described in the section 3.1.

4. EXAMPLES OF APPLICATION

4.1 Example 1

This example appeared in an article by van der Merwe and Zidek (1980), which involved physical measurements of 22 early and 11 late (male and female) respondents. The measurements of weight (y_1) and waist (y_2) are taken as the dependent variables, while the independent variables are taken as sex (x_1), height (x_2), leg length (x_3), arm length (x_4) and ID number (x_5). Although this is not an ideal example, it does serve to illustrate the prediction procedures. It is assumed that the data were selected randomly from the target population (within which prediction of the response variables is necessary).

A true assessment of a subset model is only valid if it was chosen *a priori* and not by (say) an all subset procedure on the existing data set (Miller (1984) and Copas (1983)). Hence it would be preferable to select the subset model on an entirely different set of observations to those which are to be used in the resampling procedures. In situations where this is not possible, subset models could be chosen *a priori* on as small a portion of the data as possible. In this example, the 22 early respondents and Mallows's C_p were used to select predictor variables (Table I).

TABLE I

j	Response variables	Variables selected for inclusion in $X_{in}(j)$ and hence Z_{in}	C_p	R^2
1	y_1	x_2	0,531	0,61
2	y_2	x_1 and x_2	0,390	0,55

The extensions of Mallow's C_p (see Sparks, *et al.* (1983)) were used to select the variables to be included in X_{in} of (5). These extensions ranked the subset $\{x_1, x_2\}$ as best.

4.1.1 Cross validation

The split of the data into a validation and construction sample is based partially on Miller's (1974) "Multisample jackknives". The 33 available observations are split into a construction sample of size 22 and a validation sample of size 11. As only six females responded to the request for physical measurements, a decision was taken to split females equally into the validation and construction sample. It is computationally too expensive to examine all $\begin{bmatrix} 6 \\ 3 \end{bmatrix} \times \begin{bmatrix} 27 \\ 8 \end{bmatrix}$ splits of the data into the two samples. This computational load is avoided by considering a "partial multisample jackknife" for splitting the data, established as follows:

- (a) The females were split equally into the two samples in $\begin{bmatrix} 6 \\ 3 \end{bmatrix}$ different ways.
- (b) 'Almost randomly' 20 000 different splits of the 27 males were constructed into a group of eight and another of nineteen observations.
- (c) The 20 000 splits in (b) were divided into 20 groups containing 1 000 each. Each thousand was randomly grouped with a particular split of the females to form altogether 20 000 different splits of the data into a validation sample size of 11 and a construction sample size of 22.

The standard error for prediction was estimated, using the

above 20 000 validation samples for each method. The results are summarized in table II. The performance of the gls estimator for prediction of response 1 will be identical to ols estimator using model $y_1 = \beta_{01} + x_2\beta_{21} + e_1$ where β_{21} is the regression coefficient relating to height when predicting weight. It has been proved that prediction of response 1 values using the gls estimator will be equivalent to ols prediction using the multiple regression model with predictors $X_{in}(1)$ if $X_{in}(1) \subset X_{in}(2)$ (Sparks, (1984)). Note that $\{x_2\} \subset \{x_1, x_2\}$. The cross validation results indicate that for response 1, subset model $\{x_2\}$ is better than $\{x_1, x_2\}$, which is also preferred to the full set of predictors. The benefit of the gls estimator over the ols estimator for prediction of response 2 can be assessed by comparing 2,257 and 2,190. The reduction in standard errors for prediction caused by using the gls estimator for predicting response 2 is $2,257 - 2,190 = 0,067$. Thus in this example the gls estimator seems to have the edge over other prediction procedures.

The frequencies of the rankings are expressed as percentages in table III. This table of frequencies provides additional evidence for the choice of gls estimator in predicting response 2 since out of the 20 000 splits of the data the gls estimator was ranked first in 10 495 cases (52,475% of the time). It is also noted that 77,685% of the time the gls estimator holds rank 1 or 2 and 98,235% of the time it dominates the first three ranks. For response 1, the gls estimator (or ols estimator using predictor x_2 only) is not favoured to the extent it was for response 2. In fact, this estimator only dominates the first two ranks 47,79% of the time while shrinkage on the subset model dominates these two ranks for 75,63% of the splits. The bimodel effect of the frequency distribution of the gls estimator for prediction of response 1 is caused by the fact that shrinkage estimators lead to

TABLE II

Method of estimation	Predictors included	STANDARD ERRORS					
		Response 1			Response 2		
		Cross validation	Parametric bootstrap	Nonparametric bootstrap	Cross validation	Parametric bootstrap	Nonparametric bootstrap
ols	full set	18,27	21,42	20,54	2,519	2,563	2,655
shrinkage (Zidek (1978))	full set	17,79	21,23	20,24	2,519	2,560	2,642
ols	x_1, x_2	15,79	19,27	17,26	2,257	2,334	2,175
shrinkage (Zidek (1978))	x_1, x_2	15,47	19,28	17,17	2,259	2,334	2,200
glS	response 1 - x_2	15,05	18,82	16,50	2,190	2,318	2,135
	response 2 - x_1, x_2						
RIDGE (Sclove's rule)	full set	-	21,33	-	-	2,754	-
RIDGE (H-K-B's rule)	full set	17,08	21,26	19,45	2,530	2,614	2,640

CORRELATION MATRIX FOR EXAMPLE 1

y_1	y_2	x_1	x_2	x_3	x_4	x_5
1,00	0,82	0,51	0,81	0,73	0,62	0,15
	1,00	0,71	0,70	0,60	0,57	0,09
		1,00	0,51	0,42	0,48	0,02
			1,00	0,89	0,71	0,07
				1,00	0,62	-0,04
					1,00	0,07
						1,00

TABLE III: Frequency distribution of the rankings for each re-sampling procedure.

Method of prediction	Ranks	FREQUENCIES QUOTED AS PERCENTAGES					
		Response 1			Response 2		
		Cross validation	Parametric bootstrap	Nonparametric bootstrap	Cross validation	Parametric bootstrap	Nonparametric bootstrap
ols using the full set of predictors	1(best)	0,790	4,4	4,0	0,135	7,0	2,8
	2	0,775	5,2	2,2	0,820	5,8	2,4
	3	3,505	9,2	5,2	7,585	10,0	3,8
	4	7,905	8,6	10,6	28,930	15,6	21,8
	5	9,345	18,2	21,2	19,990	21,6	26,2
	6	77,680	54,2	56,8	42,540	40,4	43,0
Zidek's (1978) shrinkage estimator using all predictors	1	1,895	1,8	1,2	0,135	3,4	0,8
	2	7,010	7,4	3,0	0,930	9,0	1,2
	3	8,360	6,0	6,8	5,200	6,4	3,8
	4	10,795	16,4	11,8	33,785	32,2	30,2
	5	68,135	51,2	55,8	50,210	37,8	52,8
	6	3,805	17,2	21,4	9,740	11,2	11,2
ols using subset model (x_1, x_2)	1	10,515	20,4	20,0	27,730	27,6	29,2
	2	34,355	39,6	41,2	25,930	29,8	33,6
	3	30,410	20,6	24,0	32,625	23,4	31,6
	4	9,925	7,4	6,4	3,345	5,6	2,6
	5	8,340	8,2	5,8	5,230	6,6	2,2
	6	6,455	3,8	2,6	5,145	7,0	0,8
Zidek's (1978) shrinkage estimator using the set of regressors (x_1, x_2)	1	36,390	24,2	21,4	16,770	19,2	12,2
	2	39,240	26,2	30,6	42,715	31,8	46,6
	3	14,345	27,8	32,2	28,055	27,4	33,4
	4	5,690	7,2	5,6	8,275	8,0	2,6
	5	3,620	7,0	4,4	3,995	7,8	2,2
	6	0,715	7,6	5,8	0,190	5,8	3,0
gls estimator using predictors: response 1 - x_2 response 2 - x_1 and x_2	1	41,500	40,8	47,6	52,475	34,8	53,2
	2	6,290	12,8	13,0	25,210	18,8	15,2
	3	35,960	26,4	23,8	20,550	25,6	24,8
	4	3,275	6,2	4,8	0,850	7,4	2,2
	5	4,015	4,0	2,6	0,665	5,6	2,2
	6	8,960	9,8	8,2	0,250	7,8	2,4
The adaptive multivariate ridge regression (H-K-B's rule)	1	8,910	8,4	5,4	2,755	8,0	1,0
	2	12,330	8,8	9,8	4,395	4,8	1,0
	3	7,420	10,0	8,2	5,985	7,0	3,4
	4	62,410	54,2	61,0	24,815	31,4	40,2
	5	6,545	11,4	10,2	19,910	21,0	14,4
	6	2,385	7,2	5,4	42,140	27,8	40,0
NUMBER OF SPLITS OR BOOTSTRAP SAMPLES		20 000	500	500	20 000	500	500

very small gains (or losses) over ols estimators. When the ols estimator using subset $\{x_1, x_2\}$ is ranked better than the gls estimator, then the shrinkage estimator using subset $\{x_1, x_2\}$ is also generally ranked above the gls. This has the effect of often pushing the gls estimator down to possible third ranked position. As such the comparison of the first two ranking positions may be unfair in this situation.

4.1.2 Nonparametric bootstrap

The nonparametric bootstrap was initially carried out by drawing randomly, with replacement, a sample of 22 observations (3 females and 19 males) from the 33 available. These were taken as the construction sample. All observations in the bootstrap population but not in the bootstrap (construction) sample are included in the validation sample. The standard errors for prediction and the frequency distribution of the rankings (expressed as percentages) are recorded in tables II and III respectively. The results are almost identical to the cross validation approach, however the standard errors are generally higher using the nonparametric bootstrap. The information gain provided by the correlated errors of the regression model can be estimated in the prediction of response 2 as $2.175 - 2.135 = 0.04$ (i.e. the difference between the standard errors of the gls and ols estimators).

To see how the bootstrap sample size affects the estimate of standard errors, various bootstrap sample sizes were considered and the standard errors were calculated for each prediction method. In the cases where the bootstrap samples were 13 and 11, problems of singularities in the estimated covariance matrix were encountered. Here a small positive ridging value was added to the zero eigenvalue of this matrix (i.e. if $S_{2 \times 2} = UD_\lambda U'$ where $D_\lambda = \begin{bmatrix} \lambda & 0 \\ 0 & 0 \end{bmatrix}$ then D_λ was replaced by $D_\lambda = \begin{bmatrix} \lambda & 0 \\ 0 & k \end{bmatrix}$ where $k = 0.00001$).

This ensured nonsingularity of the estimated variance covariance matrix. The results are summarised in table IV. The message evident from table IV is the fact that a practitioner interested in predicting accurately should concentrate on drawing as large a sample as economically possible. As such not only would the prediction errors be considerably reduced but the chances of establishing which prediction procedure is "best" would be increased. A surprising feature of table IV is the fact that even for a small sample size gls seemed to dominate other prediction methods.

4.1.3 Parametric bootstrap

Here height, weight, waist, arm length and leg length were all generated from a multivariate normal distribution. Sex variable was generated from a Bernoulli distribution with the probability of the person being female equalling the relative frequency of females in the original sample. The variable relating to registration number is generated from a discrete uniform taking on integer values between 1 and 999. The latter variable is considered a spurious variable which is uncorrelated with all other variables. The standard errors for prediction and the frequency distribution of the rankings are in table II and table III respectively. The assumption of normality seems to dampen the importance (or dominance) of the gls estimate over other prediction procedures. Although the parametric bootstrap still suggests that the gls method is "best", there is now only a slight improvement over the ols estimator using predictors (x_1, x_2) for estimating response 2 (i.e. the standard error improvement is $2,334 - 2,318 = 0,016$). This feature is also evident in the frequency distribution of the rankings, i.e. the gls estimator for prediction of response 2 is now only ranked first 34,8% of the time as opposed to 52,475% and 53,2% by the cross validation and

TABLE IV: Standard errors of prediction for 500 bootstrap samples

Method of prediction	Number of "bootstrapped" females included in the bootstrap sample	6	5	3	2	2	2	2
	Bootstrap sample size	32	27	22	17	15	13	11
		Resp. 1 Resp. 2	Resp. 1 Resp. 2	Resp. 1 Resp. 2	Resp. 1 Resp. 2	Resp. 1 Resp. 2	Resp. 1 Resp. 2	Resp. 1 Resp. 2
ols using the full set of predictors		19,06 2,415	19,75 2,490	20,54 2,655	23,05 2,987	23,74 3,041	25,36 3,17	27,52 3,423
Zidek's (1978) shrinkage estimator on the full set of predictors		18,93 2,408	19,57 2,480	20,24 2,642	22,64 2,948	23,26 3,003	24,69 3,114	26,98 3,365
ols estimate using predictors x_1, x_2		16,59 2,040	16,94 2,081	17,26 2,175	18,32 2,328	18,10 2,298	18,15 2,313	18,30 2,300
Zidek's (1987) shrinkage estimator using predictors x_1, x_2 (taking $k-q-l=1$)		16,60 2,052	16,93 2,095	17,17 2,200	18,16 2,343	18,03 2,313	18,11 2,324	18,19 2,309
gls estimator using predictors: x_2 for response 1 x_1, x_2 for response 2		16,19 2,015	16,45 2,054	16,50 2,135	16,98 2,254	16,86 2,223	17,25 2,274	17,25 2,278
Adaptive multivariate ridge using the modified Hoerl, Kennard and Baldwin rule		18,41 2,417	18,99 2,496	19,45 2,640	21,47 2,921	21,89 2,946	23,06 3,048	23,33 3,280
Total number of validations		6017	7154	8422	9799	10379	10988	11621

nonparametric bootstrap procedures respectively.

The two adaptive multivariate ridge regression rules can be compared by examining their standard errors in table II. For this example, the H-K-B's rule is preferred to Sclove's rule (i.e. $21,33 > 21,26$ and $2,754 > 2,614$).

4.2 Example 2

This example was taken from Draper and Smith (1981) page 410, exercise H. The response variables are annual figures for Total Rubber Consumption (y_1) and Tyre Rubber Consumption (y_2). The predictors are annual figures for Car Production (x_1), Gross National Product (x_2), Disposable Personal Income Tax (x_3) and Motor Fuel Consumption (x_4).

An outlier study found the observations recorded in 1950 (the data ranged from 1948 to 1963) to be influential. The studentized residuals and Cook's distance for the separate multiple regression models are recorded in table V.

TABLE V

Response	Studentized residual	Cook's distance
y_1	-2,593	3,809
y_2	-2,033	2,770

Initially prediction was assessed with all observations included. The cross validation procedure was established by considering all possible splits of the data into a construction sample of size 13 and a validation sample of size 3. The nonparametric bootstrap was conducted by taking 500 bootstrap samples of size 15 from the bootstrap population consisting of all 16 obser-

uations. The bootstrap sample formed the construction sample, while the remaining observation, not represented in the construction sample, made up the validation sample. The parametric bootstrap was conducted by assuming the response and predictor variables were normally distributed with mean vector $u' = [\bar{y}', \bar{x}']$ and covariance matrix $\Sigma = S^*$. A bootstrap sample of size $n_c + n_v = 94$ was generated from $N(u, \Sigma)$ and randomly split into a construction sample of size $n_c = 16$ and a validation sample of size $n_v = 78$. The results were obtained by generating at random such bootstrap samples. The standard errors are recorded in table VI and the frequency distribution of the rankings is recorded in table VII for all three re-sampling procedures.

Notice the peculiar frequency distribution in table VII for the gls estimator (i.e. the two largest frequencies being the first and the last rank having second highest and highest frequencies respectively). Also from table VI, it is noticeable that in terms of standard errors the gls estimator is not always the prediction method with the smallest standard errors, i.e. the parametric bootstrap chooses Zidek's shrinkage estimator for model containing predictors $\{x_1, x_2, x_3\}$ as best. The contradiction between the cross validation and the parametric bootstrap in choosing the better prediction method needs further examination. The influential observation seems to be causing some adverse effects. For this reason the cross validation program was re-run, but on this occasion only splits of the data which had the influential observation included in the validation sample was considered (this occurred 105 times in 560 splits). The results are summarized in table VIII.

Table VIII clearly indicates that the gls estimators used in this analysis are only reasonable predictors of the responses at

TABLE VI

Method of estimation	STANDARD ERRORS FOR PREDICTION					
	Response 1			Response 2		
	Cross validation	Parametric bootstrap	Nonparametric bootstrap	Cross validation	Parametric bootstrap	Nonparametric bootstrap
ols using the full set of predictors	0,0860	0,1073	0,1002	0,0867	0,1163	0,1016
Zidek's shrinkage estimator using the full model	0,0854	0,1071	0,0998	0,0858	0,1162	0,1009
ols estimator using predictors x_1, x_2, x_3	0,0797	0,1032	0,0872	0,0819	0,1110	0,0869
Zidek's shrinkage using variables x_1, x_2, x_3	0,0787	0,1031	0,0868	0,0801	0,1109	0,0861
gls estimator including predictors response 1: x_1, x_3 response 2: x_1, x_2, x_3	0,0584	0,1044	0,0816	0,0640	0,1131	0,0859
H-K-B's adaptive multivariate regression rule	0,0816	0,1064	0,0978	0,0839	0,1154	0,0995

CORRELATION MATRIX FOR EXAMPLE 2

x_1	x_2	x_3	x_4	y_1	y_2
1,00	0,47	0,42	0,37	0,64	0,56
	1,00	0,80	0,68	0,81	0,84
		1,00	0,55	0,86	0,85
			1,00	0,60	0,56
				1,00	0,98
					1,00

TABLE VII
Frequency distribution (as a percentage)

Prediction method	Ranks	Response 1			Response 2		
		Cross validation	Parametric bootstrap	Nonparametric bootstrap	Cross validation	Parametric bootstrap	Nonparametric bootstrap
ols using the full set of predictors	1	5.00	7	3.4	5.71	6	4.0
	2	6.43	11	4.2	4.46	8	4.6
	3	11.25	19	18.4	8.21	16	18.4
	4	28.57	20	19.6	21.96	22	21.4
	5	22.68	24	27.8	28.57	26	26.4
	6	26.07	19	26.6	31.07	22	25.4
Zidek's shrinkage estimator	1	3.57	5	1.8	0.89	6	2.6
	2	14.46	15	8.4	6.96	12	10.2
	3	12.68	13	17.6	12.68	15	17.2
	4	19.82	20	28.4	30.71	19	27.0
	5	40.18	35	32.8	40.18	39	30.8
	6	9.29	12	11.0	8.57	9	12.8
ols using only predictors x_1, x_2, x_3	1	28.21	22	22.4	32.50	25	25.0
	2	16.07	33	27.2	21.07	32	26.2
	3	13.21	14	20.2	11.07	16	18.2
	4	18.75	15	8.6	8.93	12	9.4
	5	16.61	10	11.0	16.07	8	12.8
	6	7.15	6	10.6	10.36	7	8.4
Zidek's shrinkage estimator using only predictors x_1, x_2, x_3	1	11.25	23	22.4	16.96	29	20.4
	2	39.11	33	38.8	45.00	32	42.2
	3	23.57	18	14.2	23.04	17	13.8
	4	13.57	9	8.8	11.25	10	12.2
	5	10.18	16	11.6	3.21	11	9.2
	6	2.32	1	4.2	0.54	1	2.2
gls estimator using predictors: response 1: x_1, x_3 response 2: x_1, x_2, x_3	1	30.89	26	38.2	32.14	23	38.6
	2	3.39	1	2.0	1.96	4	2.2
	3	21.78	17	13.0	25.36	23	12.2
	4	3.21	11	4.4	6.25	7	3.2
	5	2.86	3	1.0	2.68	1	2.6
	6	37.86	42	41.4	31.61	42	41.2
R-K-B multivariate ridge regression rule	1	20.36	17	11.0	10.54	11	8.6
	2	20.36	7	19.0	20.54	11	15.4
	3	17.68	19	17.2	19.64	14	20.2
	4	16.25	25	30.4	21.96	30	27.0
	5	7.86	12	15.6	9.29	15	18.0
	6	17.50	20	6.8	18.04	19	10.6
NUMBER OF SPLITS OR BOOTSTRAP SAMPLES		560	100	500	560	100	500

RANK 1 INDICATES RANK'S BEST

TABLE VIII
Standard errors for the influential observation included in the validation sample

Prediction method	Ranks	Frequencies		Standard errors	
		Response 1	Response 2	Response 1	Response 2
ols (full set of predictors)	1	0	1	0,1488	0,1427
	2	0	0		
	3	0	1		
	4	1	3		
	5	12	38		
	6	92	62		
Zidek's shrinkage (full set of predictors)	1	0	0	0,1462	0,1382
	2	0	2		
	3	10	26		
	4	19	52		
	5	73	22		
	6	3	3		
ols (predictors x_1, x_2, x_3)	1	0	1	0,1409	0,1397
	2	1	0		
	3	17	10		
	4	58	18		
	5	19	42		
	6	10	34		
Zidek's shrinkage (predictors x_1, x_2, x_3)	1	0	0	0,1375	0,1336
	2	33	32		
	3	62	51		
	4	9	21		
	5	1	1		
	6	0	0		
gls shrinkage (predictors: response 1: x_1, x_2 response 2: x_1, x_2, x_3)	1	105	103	0,0497	0,0626
	2	0	0		
	3	0	1		
	4	0	0		
	5	0	0		
	6	0	1		
H-K-B's rule for adaptive multivariate ridge regression	1	0	0	0,1342	0,1311
	2	71	70		
	3	16	17		
	4	18	11		
	5	0	2		
	6	0	5		

the influential data point. To clarify this, a re-run of the cross validation program was implemented, however only splits of the data which included the influential observation in the construction sample were now considered. The results of this run are reflected in table IX (together with two other means of prediction which will be motivated later). Notice (from table IX) that once the influential point is not included in the validation sample, then the gls estimator originally reflected as good in table VI is not much better than the full set of predictors. It is quite clear that the gls estimator as reflected in table VI has only a smaller standard error than the other prediction procedures because this method does well when the influential data point is included in the validation sample. To suggest an optimal prediction procedure for estimation of all data points in the situation where all 16 observations are used in estimation of parameters, is extremely difficult. However, from this analysis we have gained some idea which predictors should not be used in particular situations.

It is very important to note that if the influential point is used to estimate Σ , then from table IX, it is easy to see that all estimators which attempt to use the information in Σ perform worse than if this information is ignored.

Now the influential observation was discarded and the whole analysis was repeated. The variables selected are reflected in tables X and XI. The standard errors using only 15 observations are presented in table X and the frequency of the rankings (as percentages) are recorded in table XI. Note that the "new" gls estimator is reflected as better than the other procedures considered in table X and table XI (the "new" gls estimator is defined as the gls estimator in equation (7) using predictors x_2 , x_3 , x_4 and x_2 for response 1 and 2 respectively). To see

TABLE IX

Standard errors for cross-validation procedure
considering only splits of the data that included
the influential observation in the construction sample

	Response 1	Response 2
ols using the full set of predictors	0,0631	0,0675
Zidek's shrinkage estimator	0,0636	0,0682
ols using predictors x_1, x_2, x_3	0,0569	0,0612
ols using predictor x_2	0,0522	0,0531
Zidek's shrinkage using predictors x_1, x_2, x_3	0,0571	0,0615
glS using predictors x_1, x_3 for y_1 x_1, x_2, x_3 for y_2	0,0602	0,0644
glS using predictors x_2, x_3, x_4 for y_1 x_2 for y_2	0,0502	0,0531
H-K-B's rule using all predictors	0,0635	0,0685

TABLE X
Standard errors (observation in 1950 is excluded)

Prediction method	Response 1			Response 2		
	Cross validation	Parametric bootstrap	Nonparametric bootstrap	Cross validation	Parametric bootstrap	Nonparametric bootstrap
ols (using the full set of predictors)	0,0524	0,0865	0,0640	0,0623	0,1016	0,0792
Zidek's shrinkage estimator using all predictors	0,0530	0,0863	0,0640	0,0630	0,1010	0,0785
James-Stein shrinkage estimator using all predictors	-	-	0,0640	-	-	0,0792
ols (using predictor x_2)	0,0413	0,0783	0,0436	0,0470	0,0897	0,0492
Zidek's shrinkage estimator using predictor x_2	0,0414	0,0784	0,0437	0,0470	0,0898	0,0492
gls estimator using predictors: response 1: x_2, x_3, x_4 response 2: x_2	0,0410	0,0761	0,0428	0,0470	0,0897	0,0492
H-K-B's rule for adaptive multivariate ridge regression	0,0524	0,0862	0,0629	0,0630	0,1011	0,0778

TABLE XI
Frequency distribution on the rankings
(1950 data excluded)

Prediction method	Ranks	Response 1			Response 2		
		Cross validation	Parametric bootstrap	Nonparametric bootstrap	Cross validation	Parametric bootstrap	Nonparametric bootstrap
ols using all predictor variables	1	0.44	2	1	0.66	6	0
	2	3.30	12	7	0.44	6	1
	3	7.47	9	4	1.99	4	3
	4	30.11	28	25	27.47	20	21
	5	27.25	18	37	29.67	22	37
	6	31.43	31	26	39.78	42	38
Zidek's shrinkage using the full set of x -variables	1	1.10	4	2	1.10	6	2
	2	7.03	7	7	2.20	6	2
	3	7.47	7	9	0.22	2	0
	4	23.30	28	19	29.23	30	32
	5	41.10	40	37	53.19	45	47
	6	20.00	14	26	14.07	11	17
ols using predictor x_2 only	1	20.44	12	25	46.37	40	60*
	2	37.58	38	38	47.91	44	34
	3	20.22	24	19	3.74	3	3
	4	7.25	4	6	1.54	9	2
	5	8.79	12	6	0.44	4	1
	6	5.72	10	6	0.00	0	0
Zidek's shrinkage using predictor x_2	1	23.30	7	21	46.59	43	33
	2	32.97	32	35	2.42	1	1
	3	21.98	35	24	45.94	40	61
	4	7.69	5	5	4.62	6	3
	5	7.47	10	10	0.00	1	1
	6	6.59	11	5	0.44	9	1
gls estimator using predictors: response 1: x_2, x_3, x_4 response 2: x_2	1	45.49	71	48	46.37	43	60*
	2	9.89	3	8	47.91	40	34
	3	39.34	16	41	3.74	3	3
	4	4.18	5	2	1.54	9	2
	5	1.10	2	0	0.44	4	1
	6	0.00	3	1	0.00	0	0
H-K-B's rule for adaptive multivariate ridge	1	8.13	4	3	4.62	3	4
	2	9.45	8	3	0.66	3	2
	3	3.74	9	5	0.88	9	0
	4	27.47	30	42	33.19	32	39
	5	14.29	18	11	15.16	19	12
	6	36.92	31	36	45.49	34	43
NUMBER OF SPLITS OR BOOTSTRAP SAMPLES		455	100	100	455	100	100

* These are identical because $\{x_2\} \subset \{x_2, x_3, x_4\}$

whether the "new" gls estimator would predict response values at the influential point satisfactorily, it was decided to return to the cross validation procedure using all 16 observations. The cross validation approach was only analyzed for all splits of that data into a construction sample of size 13 and a validation sample of size 3, of which one must be the influential observation. The standard errors for response 1 and 2 for the "new" gls estimator was 0,1447 and 0,1176 respectively. Isolating predictions of the influential point revealed the fact that the gls estimator used in table VIII is far superior to the "new" gls estimator for prediction of the response values at this point (this point was excluded when estimating the parameters), i.e. $0,1447 > 0,0497$ and $0,1176 > 0,0626$. On observing table IX, it is easy to see that even if the influential observation is included in the estimation of the parameters, the "new" gls estimator is the prediction procedure which has the smallest standard errors for prediction of all other observations besides the influential one.

Hence the final conclusions are to use the "new" gls estimator (estimated with 1950's observation excluded) for prediction of future points which are within the range of the fifteen observed predictor variables (excluding the influential data point). However, if a similar situation to 1950's predictor points arises then prediction at this point can be carried out by first giving the third observation a greater weight than any other current data point and then use the gls method reflected in table VIII. (This method is recommended on the strength that it performed reasonably well for predicting the responses in 1950 even when this point was excluded from the data used for estimation of the parameters).

5. CONCLUSION

The fact that on a few occasions the shrinkage and ridge estimators (see table IX) performed worse than ols for prediction of one or both responses suggests that mechanical application of these estimators to improve on prediction using $\hat{\beta}_0$ must be condemned (see Draper and van Nostrand (1979)). The ridge regression and shrinkage techniques enable one to incorporate prior information of a specific kind, and this is possibly the prime beneficial use of these techniques.

The empirical selection techniques based on the same data used in estimation of the β 's can lead to the selection of subset models which are far from best (see example 2; outlier included) or even worse than the full set of predictors. The resampling procedures are therefore diagnostic tools for largely guarding against this tendency. In particular, this will be true, if the variables are selected from an entirely different set of observations to those used in the resampling procedures. To quote Snee (see comments on Hocking's (1982) paper): "I am confident that if more developers of regression techniques had used model validation techniques, we would have recognized more quickly the inability of methods based on least squares (e.g. stepwise and best subsets) to adequately handle collinearity effects (Marquardt and Snee (1975))". This quote does not seem to hold true in at least the first example considered and to a lesser degree (once the influential observation was removed) in the second example. However, selection of variables will not always reduce the effects of collinearity.

If the resampling techniques are carried out as is proposed in this paper then I believe practitioners may be going a long way towards finding the better of rival prediction methods. Also

these techniques may be very useful diagnostic tools for: avoiding the use of poor subset models for prediction, detection of influential and outlying points, checking how the inclusion of these points affect prediction and also establishing which method will best predict the response values at the outlying points. The proposed approach is not intended as a substitute for other regression diagnostics but to complement existing techniques.

The generalised least squares estimator as has been shown to be a useful technique for estimating regression coefficients in seemingly unrelated regression equations (Swamy and Mehta (1976) and Sparks (1987)) may prove to be beneficial for prediction in multivariate regression models.

If a "smoothed" histogram (of the validation sum of squares) is constructed for each method of prediction, and these are all plotted on the same axes using colour graphics, then this could be a particularly useful decision making and diagnostic tool. If this was done for the cross validation and nonparametric bootstrap procedures then any multimodality of the empirical distributions would indicate an influential or an outlying observation (or a clump of these observations).

The suggested resampling procedures as prescribed in this paper can be used to compare prediction procedures for a much larger class of models than just the multivariate regression model. It could, for example, be used to choose between the gls and ols estimator of the regression coefficients from prediction in the seemingly unrelated regression model (Sparks (1987)) when the errors are (say) not normally distributed.

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